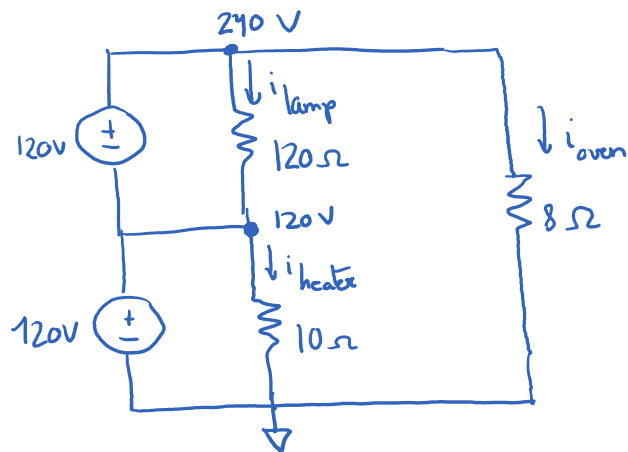


Q1a



* Lamp

$V_{\text{lamp}} = 120\text{V}$ (since it is in parallel with the source)

$$i_{\text{lamp}} = \frac{V_{\text{lamp}}}{R_{\text{lamp}}} = \frac{120}{120} = \boxed{1\text{A}}$$

* Heater (same situation as lamp)

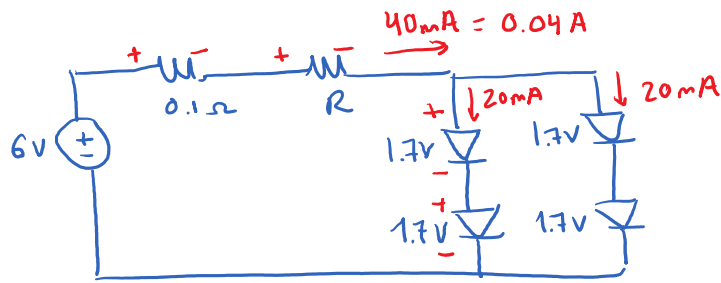
$$i_{\text{heater}} = \frac{V_{\text{heater}}}{R_{\text{heater}}} = \frac{120}{10} = \boxed{12\text{A}}$$

* Oven

$V_{\text{oven}} = 240\text{V}$ (as per nodal voltages)

$$i_{\text{oven}} = \frac{V_{\text{oven}}}{R_{\text{oven}}} = \frac{240}{8} = \boxed{30\text{A}}$$

Q1b

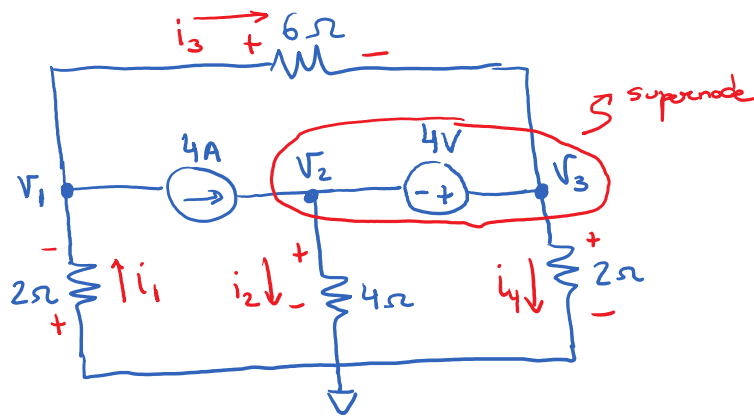


$$i_{\max \text{ LED}} = 20\text{mA}$$

- * If we want 20mA in each of the LED branches, we need $20 + 20 = 40\text{mA}$ (KCL) in the left mesh.
- * Use KVL to find voltage across R :

$$-6 + 0.04 \times 0.1 + 0.04 R + 1.7 + 1.7 = 0$$

$$0.04 R = 6 - 1.7 - 1.7 - 0.004 \Rightarrow \boxed{R = \frac{2.596}{0.04} = 64.9\Omega}$$



* KCL @ node 1: $i_1 - 4 - i_3 = 0$

$$\frac{0 - V_1}{2} - 4 - \frac{V_1 - V_3}{6} = 0$$

* KCL @ supernode:

$$4 - i_2 - i_4 + i_3 = 0$$

$$4 - \frac{V_2 - 0}{4} - \frac{V_3 - 0}{2} + \frac{V_1 - V_3}{6} = 0$$

$$\Rightarrow 48 - 3V_2 - 6V_3 + 2V_1 - 2V_3 = 0$$

* Constraint equation:

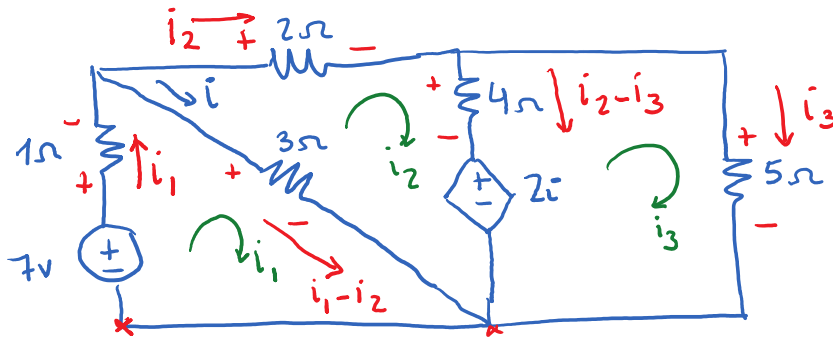
$$V_3 - V_2 = 4$$

$$\Rightarrow 2V_1 - 3V_2 - 8V_3 = -48$$

Equations are:

$$\begin{cases} -4V_1 + V_3 = 24 \\ 2V_1 - 3V_2 - 8V_3 = -48 \\ V_2 + V_3 = 4 \end{cases}$$

Q2b



* KVL @ mesh 1:

$$-7 + 1i_1 + 3(i_1 - i_2) = 0 \Rightarrow 4i_1 - 3i_2 = 7$$

* KVL @ mesh 2:

$$-3(i_1 - i_2) + 2i_2 + 4(i_2 - i_3) + 2i, \text{ where } \boxed{i = i_1 - i_2}$$

$$-3i_1 + 3i_2 + 2i_2 + 4i_2 - 4i_3 + 2i_1 - 2i_2 = 0 \Rightarrow -i_1 + 7i_2 - 4i_3 = 0$$

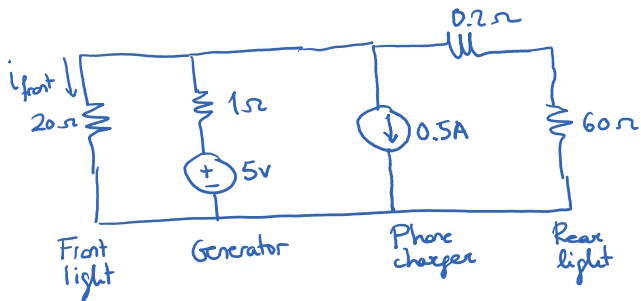
* KVL @ mesh 3:

$$-2i - 4(i_2 - i_3) + 5i_3 = 0, \text{ where } i = i_1 - i_2$$

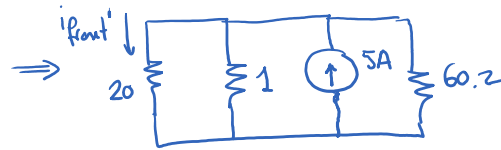
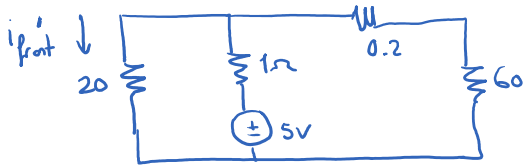
$$-2i_1 + 2i_2 - 4i_2 + 4i_3 + 5i_3 = 0 \Rightarrow -2i_1 - 2i_2 + 9i_3 = 0$$

Equations are:

$$\boxed{\begin{aligned} 4i_1 - 3i_2 &= 7 \\ -i_1 + 7i_2 - 4i_3 &= 0 \\ -2i_1 - 2i_2 + 9i_3 &= 0 \end{aligned}}$$



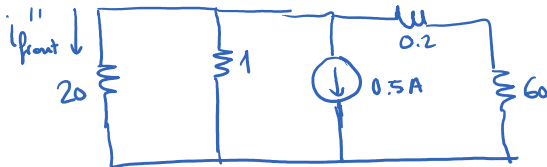
2) * Contribution of voltage source (turn off current source)



$$\boxed{i'_{\text{front}}} = 5 \frac{20 \parallel 1 \parallel 60.2}{20} = 5 \frac{0.9375}{20} = \underline{0.234 \text{ A}}$$

↑
current divider

* Contribution of current source (turn off voltage source)



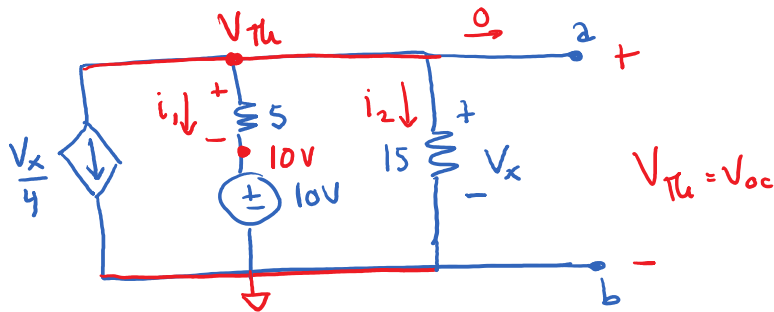
$$\boxed{i''_{\text{front}}} = -0.5 \frac{20 \parallel 1 \parallel 60.2}{20} = -0.5 \frac{0.9375}{20} = \underline{-0.0234 \text{ A}}$$

↑
current divider

* Add contributions

$$\boxed{i_{\text{front}}} = i'_{\text{front}} + i''_{\text{front}} = 0.234 - 0.0234 = \underline{0.21 \text{ A}}$$

b) $\boxed{P_{\text{front}}} = i_{\text{front}}^2 \times 20 = 0.21^2 \times 20 = \underline{0.882 \text{ W}}$



2) Calculate V_{th} :

$$V_{th} = V_{oc} = V_{15\Omega} = V_x$$

Using nodal:

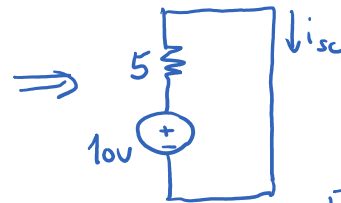
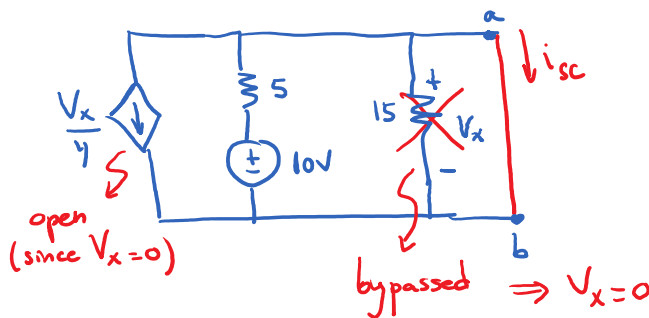
KCL @ top node: $-\frac{V_x}{4} - i_1 - i_2 = 0$, where $V_x = V_{th}$

$$-\frac{V_{th}}{4} - \frac{V_{th} - 10}{5} - \frac{V_{th} - 0}{15} = 0$$

$$-15V_{th} - 12V_{th} + 120 - 4V_{th} = 0$$

$$-31V_{th} = -120 \Rightarrow \boxed{V_{th} = \frac{120}{31} = 3.87 \text{ V}}$$

b) Calculate R_{th} (method 2)



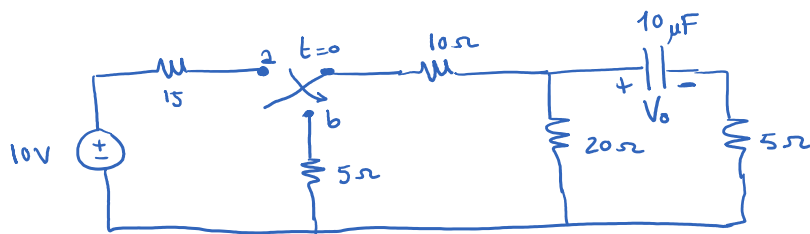
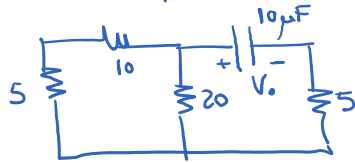
$$i_{sc} = \frac{10\text{V}}{5\Omega} = 2\text{A}$$

Ohm's law

$$\boxed{R_{th} = \frac{V_{th}}{i_{sc}} = \frac{3.87}{2} = 1.94\Omega}$$

c) Calculate max. power

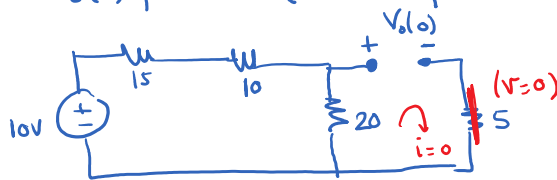
$$\boxed{P_{max} = \frac{V_{th}^2}{4R_{th}} = \frac{3.87^2}{4 \times 1.94} = 1.93 \text{ W}}$$

Find $V_o(t)$ for $t > 0$ For $t > 0$ switch is in position b

Natural response

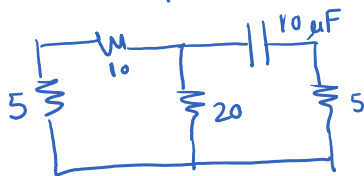
$$V_o(t) = V_o(0) e^{-t/\tau} \quad \text{where } \tau = R_{eq} \times C$$

* Calculate $V_o(0)$ for $t < 0$ (switch in position a) $\Rightarrow$ steady state
 $\hookrightarrow$ capacitor is open circuit



$$\Rightarrow V_o(0) = V_{20\Omega} = 10 \frac{20}{15+10+20} = 4.444 \text{ V}$$

voltage divider

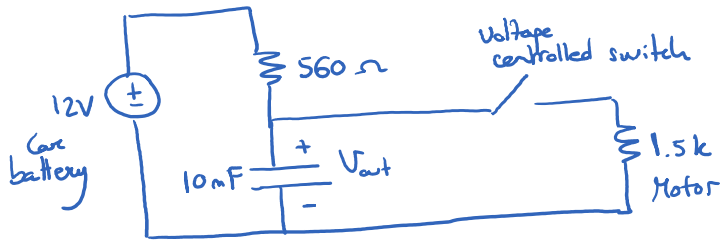
* Calculate τ for $t > 0$ 

$$\tau = R_{eq} \times C = 13.57 \times 10 \mu\text{F} = 135.7 \mu\text{s}$$

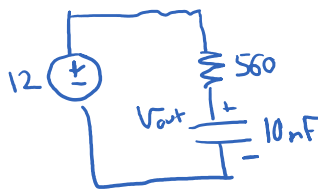
$$R_{eq} = [(5+10) \parallel 20] + 5 = 13.57 \Omega$$

* Substitute:

$$\boxed{V_o(t) = V_o(0) e^{-t/\tau} = 4.444 e^{-t/0.0001357} = 4.444 e^{-7368.4 t} \text{ V}}$$



i)

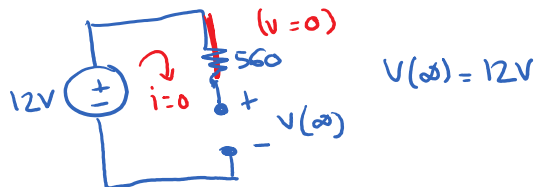


Switch is open and we want to know how long does it take for the capacitor to reach 10V.

Step response: $V_{out}(t) = V(\infty) + [V(0) - V(\infty)]e^{-t/\tau}$
where $\tau = RC$

* $V(0) = 0V$

* Calculate $V(\infty)$:
capacitor in steady state $\Rightarrow$ open circuit



* Calculate τ : $\tau = RC = 560 \times 10 \text{ mF} = 5.6 \text{ s}$

* Substitute: $V_{out}(t) = 12 + [0 - 12]e^{-t/5.6} = 12(1 - e^{-t/5.6})$

* Make $V_{out} = 10V$ and find t

$$12(1 - e^{-t/5.6}) = 10$$

$$1 - e^{-t/5.6} = 10/12$$

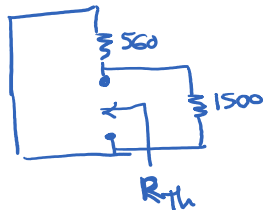
$$+e^{-t/5.6} = \frac{10}{12} - 1 = -0.1667$$

$$+t/5.6 = \ln(0.1667) = -1.792$$

$$\boxed{t = 1.792 \times 5.6 = 10 \text{ s}}$$

ii)

* $\tau = R_{th} \times C$



$$R_{th} = 1500 \parallel 560 = 407.77$$

$$\tau = R_{th} \times C = 407.77 \times 10 \text{ mF} = 4.0777$$